

Contracting $T^{mm'}$ with the bracket

Every index step, and the final cross section

$$S = p^+ + k^+, \quad \zeta = \frac{k^+}{S}, \quad \bar{\zeta} = \frac{p^+}{S} = 1 - \zeta, \quad r = y' - y, \quad \hat{r} = r/|r|, \quad d_\perp = 2 - 2\epsilon$$

0. The three objects being contracted

After the w, w' integration and the Wilson–line reduction, the transverse integrand is

$$\begin{aligned} & \underbrace{\frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2}}_{\rightarrow T^{mm'} \text{ after } \int d^d z} \times \underbrace{\frac{(x-y)^k}{(x-y)^2}}_{X^k} \underbrace{\frac{(x'-y')^{k'}}{(x'-y')^2}}_{X'^{k'}} \times \mathcal{B}_{mkm'k'} \\ & \mathcal{B}_{mkm'k'} = b_1 \delta_{km} \delta_{k'm'} + b_2 \delta_{mm'} \delta_{kk'} + b_3 \delta_{mk'} \delta_{km'} \\ & b_1 = \frac{d_\perp p^+}{S^2} - \frac{2}{k^+}, \quad b_2 = \frac{p^+}{k^+2} + \frac{1}{p^+}, \quad b_3 = \frac{2}{k^+} \\ & T^{mm'} = C_\epsilon \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad \frac{C_\epsilon}{\epsilon} \rightarrow \pi(\mathcal{L} + 2), \quad C_\epsilon \rightarrow \pi \end{aligned}$$

Call the whole contraction $\mathcal{Q} \equiv T^{mm'} X^k X'^{k'} \mathcal{B}_{mkm'k'}$. Four indices, three terms — do them one at a time.

1. The three index contractions, one at a time

1.1 The b_1 term: $\delta_{km} \delta_{k'm'} T^{mm'} X^k X'^{k'}$

δ_{km} is summed over k and replaces k by m wherever k appears, i.e. in X^k . $\delta_{k'm'}$ does the same for k' in $X'^{k'}$. Nothing else carries k or k' . So

$$\delta_{km} \delta_{k'm'} T^{mm'} X^k X'^{k'} = T^{mm'} X^m X'^{m'} \equiv T(X, X')$$

1.2 The b_2 term: $\delta_{mm'} \delta_{kk'} T^{mm'} X^k X'^{k'}$

Here the two deltas act on *separate* pairs: $\delta_{mm'}$ ties T 's own two indices together, $\delta_{kk'}$ ties X to X' . The two factors therefore decouple:

$$\delta_{mm'} \delta_{kk'} T^{mm'} X^k X'^{k'} = (\delta_{mm'} T^{mm'}) (X^k X'^k) = (\text{tr } T) \mathcal{X}, \quad \mathcal{X} \equiv X \cdot X'$$

1.3 The b_3 term: $\delta_{mk'} \delta_{km'} T^{mm'} X^k X'^{k'}$

This one crosses the indices. $\delta_{mk'}$ replaces k' by m in $X'^{k'}$; $\delta_{km'}$ replaces k by m' in X^k :

$$\delta_{mk'} \delta_{km'} T^{mm'} X^k X'^{k'} = T^{mm'} X'^m X^{m'} \equiv T(X', X)$$

So X and X' have swapped which slot of T they sit in.

1.4 But T is symmetric, so 1.3 equals 1.1

Recall the symmetry proof: in $T^{mm'} = \int d^d u \frac{u^m (u+r)^{m'}}{u^2 (u+r)^2}$ send $u \rightarrow -u - r$. The measure is invariant, both denominators are invariant, and the numerator becomes $(u+r)^m u^{m'}$ — which is the original with $m \leftrightarrow m'$. Hence $T^{mm'} = T^{m'm}$, and

$$T(X', X) = T^{mm'} X'^m X^{m'} = T^{m'm} X'^m X^{m'} = T(X, X').$$

structure	reduces to	numeric	closed form
$\delta_{km} \delta_{k'm'}$	$T(X, X')$	0.494031134	0.494031134
$\delta_{mm'} \delta_{kk'}$	$(\text{tr } T) \mathcal{X}$	0.006421621	0.006421621
$\delta_{mk'} \delta_{km'}$	$T(X', X) = T(X, X')$	0.494031134	0.494031134

$$\mathcal{Q} = (b_1 + b_3) T(X, X') + b_2 (\text{tr } T) \mathcal{X}$$

Two structures, not three — because T is symmetric, the first and third of your bracket's tensors collapse onto the same thing.

2. $b_1 + b_3$: the $\frac{2}{k^+}$ cancels, and that is the momentum sum rule

Trace the $\frac{2}{k^+}$ back to where it came from. Before the w integration the three coefficients were

$$c_1 = d_\perp - \frac{2S}{p^+} - \frac{2S}{k^+}, \quad c_2 = S^2 \left[\frac{1}{p^{+2}} + \frac{1}{k^{+2}} \right], \quad c_3 = \frac{2S^2}{p^+ k^+}$$

and $b_i = \frac{p^+}{S^2} c_i$. So

$$b_1 = \frac{d_\perp p^+}{S^2} - \underbrace{\frac{2}{S}}_{\frac{2\zeta}{k^+}} - \underbrace{\frac{2p^+}{S k^+}}_{\frac{2\bar{\zeta}}{k^+}} = \frac{d_\perp p^+}{S^2} - \frac{2}{k^+} (\zeta + \bar{\zeta}) = \frac{d_\perp p^+}{S^2} - \frac{2}{k^+}$$

because $\zeta + \bar{\zeta} = 1$. And $b_3 = \frac{p^+}{S^2} \frac{2S^2}{p^+ k^+} = \frac{2}{k^+}$ exactly. Therefore

$$\boxed{b_1 + b_3 = \frac{d_\perp p^+}{S^2}} \quad (\text{symbolic check: difference } 0)$$

The cancellation is not an accident of algebra. $\frac{2}{S}$ came from the $\frac{S}{p^+}$ structure of the $\mathfrak{C}$ vertex and $\frac{2p^+}{S k^+}$ from the $\frac{S}{k^+}$ one; they add to $\frac{2}{k^+} (\zeta + \bar{\zeta})$, and $\zeta + \bar{\zeta} = 1$ **is momentum conservation at the splitting vertex**. That is what lets the b_3 term cancel it cleanly. If you ever find this *not* cancelling, you have made a kinematic error upstream.

3. The two T -contractions in $d_\perp = 2 - 2\epsilon$

3.1 $T(X, X')$ — *nothing is eaten*

$$T(X, X') = X_m X'_{m'} C_\epsilon \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right] = C_\epsilon \left[\frac{X \cdot X'}{2\epsilon} - (X \cdot \hat{r})(X' \cdot \hat{r}) \right] = C_\epsilon \left[\frac{\mathcal{X}}{2\epsilon} - \mathcal{R} \right]$$

X and X' are ordinary transverse vectors built from x, y, x', y' — they carry **no** ϵ . So the $\frac{1}{\epsilon}$ pole meets nothing that could cancel it, and $C_\epsilon \left[\frac{1}{2\epsilon} \right] \rightarrow \frac{\pi}{2} (\mathcal{L} + 2)$ intact:

$$T(X, X') \longrightarrow \frac{\pi}{2} (\mathcal{L} + 2) \mathcal{X} - \pi \mathcal{R}$$

3.2 $\text{tr } T$ — *here the $+2$ is eaten*

$$\text{tr } T = \delta_{mm'} C_\epsilon \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right] = C_\epsilon \left[\frac{\delta_{mm}}{2\epsilon} - \hat{r} \cdot \hat{r} \right] = C_\epsilon \left[\frac{d_\perp}{2\epsilon} - 1 \right]$$

Two things to be careful about: $\delta_{mm} = d_\perp = 2 - 2\epsilon$ (this is the whole point of dimensional regularisation), while $\hat{r} \cdot \hat{r} = 1$ exactly, because $\hat{r}$ is a unit vector by construction and carries no ϵ . Then

$$\begin{aligned} \frac{d_\perp}{2\epsilon} = \frac{2 - 2\epsilon}{2\epsilon} = \frac{1}{\epsilon} - 1 &\implies \text{tr } T = C_\epsilon \left[\frac{1}{\epsilon} - 2 \right] = \frac{C_\epsilon}{\epsilon} - 2C_\epsilon \\ &\longrightarrow \pi(\mathcal{L} + 2) - 2\pi = \boxed{\pi \mathcal{L}} \end{aligned}$$

The -2ϵ in $d_\perp$ struck the $\frac{1}{\epsilon}$ and left a finite $-\pi$; together with the $-C_\epsilon$ from the $\hat{r} \hat{r}$ term it removes the $+2$ entirely. **Setting $d_\perp = 2$ before this step would have given $\pi(\mathcal{L} + 2)$ — wrong by 2π .**

4. Assemble

$$\mathcal{Q} = \frac{d_{\perp} p^+}{S^2} C_{\epsilon} \left[\frac{\mathcal{X}}{2\epsilon} - \mathcal{R} \right] + \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right) \pi \mathcal{L} \mathcal{X}$$

The first term also carries an explicit $d_{\perp}$, so it gets its own partial eating:

$$\frac{(2-2\epsilon)p^+}{S^2} C_{\epsilon} \left[\frac{\mathcal{X}}{2\epsilon} - \mathcal{R} \right] = \frac{p^+}{S^2} \left[\underbrace{(1-\epsilon) \frac{C_{\epsilon}}{\epsilon}}_{\frac{C_{\epsilon}}{\epsilon} - C_{\epsilon} \rightarrow \pi(\mathcal{L}+1)} \mathcal{X} - \underbrace{(2-2\epsilon)C_{\epsilon}\mathcal{R}}_{\rightarrow 2\pi} \right] = \frac{\pi p^+}{S^2} [(\mathcal{L}+1)\mathcal{X} - 2\mathcal{R}]$$

One unit of the $+2$ is eaten here, leaving $\mathcal{L} + 1$. So

$$\mathcal{Q} = \frac{\pi p^+}{S^2} [(\mathcal{L}+1)\mathcal{X} - 2\mathcal{R}] + \pi \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right) \mathcal{L} \mathcal{X}$$

5. To ζ variables — and P_{gg} appears

With $S = \frac{k^+}{\zeta}$ and $p^+ = \bar{\zeta} S = \frac{\bar{\zeta}}{\zeta} k^+$:

$$\frac{p^+}{S^2} = \frac{\bar{\zeta}}{S} = \frac{\zeta \bar{\zeta}}{k^+}, \quad \frac{p^+}{k^{+2}} = \frac{\bar{\zeta}}{\zeta k^+}, \quad \frac{1}{p^+} = \frac{\zeta}{\bar{\zeta} k^+}$$

$$\mathcal{Q} = \frac{\pi}{k^+} \left\{ \zeta \bar{\zeta} [(\mathcal{L}+1)\mathcal{X} - 2\mathcal{R}] + \left(\frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right) \mathcal{L} \mathcal{X} \right\}$$

Now split $\zeta \bar{\zeta} (\mathcal{L}+1)\mathcal{X} = \zeta \bar{\zeta} \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} \mathcal{X}$ and collect the three $\mathcal{L} \mathcal{X}$ pieces:

$$\mathcal{Q} = \frac{\pi}{k^+} \left\{ \underbrace{\left[\frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta \bar{\zeta} \right]}_{C_{UV}(\zeta) = P_{gg}(\zeta)/2N_c} \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}$$

Symbolic difference against the direct expansion: **exactly** 0.

Notice what supplied what:

- $\frac{\zeta}{\bar{\zeta}}$ came from b_2 's $\frac{1}{p^+}$ — the **soft** end;
- $\frac{\bar{\zeta}}{\zeta}$ came from b_2 's $\frac{p^+}{k^{+2}}$;
- $\zeta \bar{\zeta}$ — the **regular** part of P_{gg} , the piece no soft approximation can produce — came from $b_1 + b_3$, i.e. from the $d_{\perp}$ term after the $\frac{2}{k^+}$ cancellation;
- and the leftover "+1" from §4 is exactly what supplies the finite $\zeta \bar{\zeta} \mathcal{X}$.

6. The final cross section

Restoring the prefactor $\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{N_c}{k^+}$, the $\frac{dp^+}{2\pi}$, and $\mathcal{Q}$'s own $\frac{\pi}{k^+}$:

$$\frac{1}{(2\pi)^3} \cdot \frac{g^4}{4\pi^4} \cdot \frac{N_c}{k^+} \cdot \frac{1}{2\pi} \cdot \frac{\pi}{k^+} = \frac{g^4 N_c}{64\pi^7 k^{+2}}$$

$$\frac{d^3 N_{\text{NLO}}}{d^3 k} = \frac{g^4 N_c}{64\pi^7 k^{+2}} \int_{\Lambda}^{\vee - k^+} dp^+ \int_{x, x', y, y'} e^{-i \frac{k}{\zeta} \cdot (y' - y)} \left\{ C_{\text{UV}}(\zeta) \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\} \\ \times \left\langle [U(y') - U(x')]^{e'c} [U(y) - U(x)]^{ec} \rho^{e'}(x') \rho^e(x) \right\rangle$$

with everything written out:

$$\mathcal{X} = \frac{(x - y) \cdot (x' - y')}{(x - y)^2 (x' - y')^2}, \quad \mathcal{R} = \frac{[(x - y) \cdot (y' - y)] [(x' - y') \cdot (y' - y)]}{(x - y)^2 (x' - y')^2 (y' - y)^2} \\ C_{\text{UV}}(\zeta) = \frac{\zeta}{1 - \zeta} + \frac{1 - \zeta}{\zeta} + \zeta(1 - \zeta) = \frac{P_{gg}(\zeta)}{2N_c}, \quad \zeta = \frac{k^+}{p^+ + k^+} \\ \mathcal{L} = \frac{1}{\epsilon_{\overline{\text{MS}}}} - \log(4\pi^2 \mu^2 (y' - y)^2), \quad \frac{1}{\epsilon_{\overline{\text{MS}}}} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi$$

6.1 The same thing in ζ — the fragmentation form

$$p^+ = k^+ \frac{1 - \zeta}{\zeta} \implies |dp^+| = \frac{k^+}{\zeta^2} d\zeta, \quad p^+ = \Lambda \rightarrow \zeta = \frac{k^+}{k^+ + \Lambda}, \quad p^+ = \vee - k^+ \rightarrow \zeta = \frac{k^+}{\vee}$$

$$\frac{d^3 N_{\text{NLO}}}{d^3 k} = \frac{g^4 N_c}{64\pi^7 k^+} \int_{k^+/\vee}^{k^+/(k^+ + \Lambda)} \frac{d\zeta}{\zeta^2} \int_{x, x', y, y'} e^{-i \frac{k}{\zeta} \cdot (y' - y)} \left\{ C_{\text{UV}}(\zeta) \mathcal{L} \mathcal{X} + \zeta(1 - \zeta) [\mathcal{X} - 2\mathcal{R}] \right\} \langle \dots \rangle$$

All three fragmentation hallmarks are now manifest: the measure $\frac{d\zeta}{\zeta^2}$, the hard factor at the parent momentum $k_{\perp}/\zeta$, and $P_{gg}(\zeta)$ multiplying $\log \mu^2$.

6.2 After $\overline{\text{MS}}$ and the $+$ prescription

Absorb $\frac{1}{\epsilon_{\overline{\text{MS}}}}$ into the gluon fragmentation function $D_{g/g}(\zeta, \mu)$, so $\mathcal{L} \rightarrow -\log(4\pi^2 \mu^2 (y' - y)^2)$. Then isolate the $\zeta \rightarrow 1$ endpoint, the only singular piece of C_{UV} :

$$C_{\text{UV}}(\zeta) \longrightarrow \zeta \left[\frac{1}{1 - \zeta} \right]_+ + \frac{1 - \zeta}{\zeta} + \zeta(1 - \zeta) + \delta\text{-term}$$

where, with $a = \frac{k^+}{\vee}$ and $\delta = \frac{\Lambda}{k^+}$, the δ -term is the explicit

$$F(1) \log \frac{k^+ (\vee - k^+)}{\vee \Lambda}, \quad F(1) = e^{-ik \cdot (y' - y)} \mathcal{L} \mathcal{X} \text{ (the LO structure),}$$

which is the piece that goes into the small- x evolution. Everything else is finite and Λ -independent.

7. Summary of the logic

1. T is symmetric $\implies$ the $\delta_{km} \delta_{k'm'}$ and $\delta_{mk'} \delta_{km'}$ structures collapse onto one, so b_1 and b_3 add.
2. $b_1 + b_3 = \frac{d_{\perp} p^+}{S^2}$: the $\frac{2}{k^+}$ cancels by $\zeta + \bar{\zeta} = 1$.
3. $T(X, X')$ keeps its $\mathcal{L} + 2$; $\text{tr } T$ loses it entirely and becomes $\pi \mathcal{L}$; the $d_{\perp}$ in $b_1 + b_3$ eats one unit, giving $\mathcal{L} + 1$.
4. In ζ variables the three surviving $\mathcal{L} \mathcal{X}$ coefficients assemble into exactly $C_{\text{UV}}(\zeta) = P_{gg}/2N_c$, with $\zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}]$ as the finite remainder.

Every step checked in `src/row24_contract.py`; output in `RESULTS_row24.txt`.